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Inventor(s)

HOCH; Charles V. et al.

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### MOUNTING SYSTEM FOR SECURING ITEMS TO VEHICLE

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#### Abstract

A mounting system for removably attaching components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting system including a mounting assembly. The mounting assembly is configured to be secured to the sidewall and to support a load; a load support structure, wherein the load support structure is configured to couple to the mounting assembly; an engagement between the mounting assembly and the load support structure being positioned at or below the top rim of the sidewall.

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**Inventors:** HOCH; Charles V. (Durango, CO), HOCH; Bruce A. (Durango, CO)

**Applicant:** HOCH BRANDS LLC (Boca Raton, FL)

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. non-provisional application Ser. No. 17/884,957, filed 10 Aug. 2022, entitled MOUNTING SYSTEM FOR SECURING ITEMS TO VEHICLE, which claims the benefit of priority pursuant to 35 U.S.C. § 119(e) of U.S. provisional patent application No. 63/232,097, filed 11 Aug. 2021, which are each hereby incorporated by reference herein in their entirety.

### FIELD

[0002] This disclosure relates generally to mounting systems and methods for securing items to vehicles.

### BACKGROUND

[0003] Existing methods and systems of securing cargo to vehicles, such as pickup trucks, have numerous deficiencies. For example roof racks are hard to reach. Hitch mount racks interfere with the use of the tailgate and place cargo in the path of mud and dirt kicked up by the vehicle. Improved solutions are desired that can enable easy access to cargo and to the tailgate and doors of the vehicle.

### BRIEF SUMMARY

[0004] In some examples, a mounting system for removably attaching components to a vehicle including at least one sidewall defining a top rim, the mounting system includes a mounting assembly, wherein the mounting assembly is configured to be secured to the sidewall and to support a load; a load support structure, wherein the load support structure is configured to couple to the mounting assembly; and an engagement between the mounting assembly and the load support structure being positioned at or below the top rim of the sidewall. Additionally or alternatively, the vehicle may include a bed portion, where the bed portion includes the at least one sidewall defining a top rim.

[0005] In addition or alternatively, the mounting assembly may include a bracket, the bracket including a top portion, wherein the top portion is configured to be secured to the sidewall, a lower portion, wherein the lower portion defines one or more receiving portions, the one or more receiving portions defining one or more features to support one or more loads or load support structures, and an intersection portion, wherein the intersection portion defines a location where the top portion and lower portions intersect.

[0006] Additionally or alternatively, the lower portion of the mounting assembly may comprises a brace member defining a supporting feature extending from the lower portion in a direction of the sidewall.

[0007] Alternatively or additionally, the intersection portion may be reinforced by a structure, the structure spanning from the top portion to the lower portion of the bracket.

[0008] Alternatively or additionally, the intersection portion is adjustable such that a position of the lower portion may be repositioned relative to a position of the top portion.

[0009] Additionally or alternatively, the mounting assembly may also include an adjustment mechanism, the adjustment mechanism including a top plate defining at least one adjustment aperture, a bottom plate configured to define at least one corresponding adjustment aperture, wherein the top portion of the bracket is configured to define at least one corresponding adjustment aperture, and an elongated recess configured to receive the top portion of the bracket, wherein by orienting the at least one adjustment aperture of the top portion of the bracket with the at least one adjustment aperture of the bottom plate and inserting at least one fastener, the bracket may be placed in at least one position relative to the sidewall. Additionally or alternatively, the at least one adjustment aperture of the top portion of the bracket is laterally spaced away from the sidewall. Additionally or alternatively, a portion of a topper is positioned on the sidewall, and wherein the at

least one adjustment aperture is accessible.

[0010] Additionally or alternatively, the top portion is configured to define apertures, the apertures configured to receive fasteners, the fasteners configured to secure the bracket directly to the sidewall.

[0011] Additionally or alternatively, the load support structure defines one or more utility portions, the utility portion configured to support a load.

[0012] Additionally or alternatively, the load support structure further includes one or more utility portions, the utility portions configured to support a load, and one or more coupling portions, the coupling portions configured to mate with the receiving portion of one or more brackets.

[0013] Alternatively or additionally, the load support structure spans between two or more mounting assemblies.

[0014] Additionally or alternatively, the mounting assembly may include a securement mechanism, wherein the securement mechanism is configured to secure the mounting assembly to the sidewall.

[0015] Additionally, the securement mechanism is configured to removably couple to the sidewall, the securement mechanism further comprising a slider member, the slider member configured to rest in apertures defined by the top portion of the bracket, the slider member further configured to define an aperture; a clamp member; the clamp member configured to couple to a feature on the sidewall; the clamp member further defining an aperture; and a rod, wherein the rod is configured to extend through the clamp member and into the slider member, and the rod causes the slider member to exert force on the top portion of the bracket, and the clamp member to exert force on the sidewall, thereby securing the mounting assembly to the sidewall.

[0016] Alternatively or additionally, wherein the mounting assembly further includes an adjustment mechanism, the adjustment mechanism may include the slider member, the slider member configured to rest within apertures defined by the top portion of the bracket, the apertures configured to define tracks, the tracks configured to allow the slider member to rest in multiple positions within the tracks, and wherein a location of the bracket may be adjusted relative to the sidewall and secured into a location when the slider member exerts an applied force on the top portion of the bracket through the tracks.

[0017] Alternatively or additionally, the securement mechanism may include a multi-layer assembly, the multi-layer assembly including a bottom plate configured to rest on the rim of the sidewall and to define one or more securement apertures, a top plate configured to define corresponding securement apertures, a recess between the top plate and the bottom plate, the recess configured to receive the top portion of the bracket, and one or more adjustment apertures, the adjustment apertures configured to be placed above the recess, and wherein fasteners inserted into the securement apertures couple the multi-layer assembly to the sidewall, and fasteners inserted into the one or more adjustment apertures secure the bracket within the multi-layer assembly.

[0018] Additionally or alternatively a plurality of adjustment apertures may be included, the plurality of adjustment apertures configured to secure the bracket in multiple positions relative to the sidewall.

[0019] Additionally or alternatively, the mounting assembly may also include an adjustment mechanism, the adjustment mechanism including a slider member, configured to rest within apertures defined by the top portion of the bracket, the apertures configured to define tracks, the tracks configured to allow the slider member to rest in multiple positions within the tracks, and wherein a placement of the bracket may be adjusted relative to the sidewall and secured into the location when the slider member exerts an applied force on the top portion of the bracket through the tracks.

[0020] Additionally or alternatively, the tracks are further configured to define teeth, the teeth configured to be structures that define discrete locations for the slider member to rest.

[0021] Additionally or alternatively, the slider member further includes features that increase the applied force on the top portion of the bracket.

[0022] In another example, a mounting system for removably attaching utility components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting system includes a mounting assembly including a bracket, a securement mechanism, and an adjustment mechanism, the securement mechanism securing the mounting assembly to the sidewall, the adjustment mechanism allowing positioning of the mounting assembly relative to the sidewall; a load support structure including one or more utility portions and one or more coupling portions, the coupling portion for removably connecting the fixture to the bracket of the mounting assembly; and the engagement between the bracket and the coupling portion being positioned at or below the top rim of the sidewall.

[0023] In another example, the mounting system for removably attaching components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting system including a mounting assembly including a bracket, and a securement mechanism, the bracket configured to support a load, the securement mechanism securing the mounting assembly to the sidewall, and the engagement between the mounting assembly and the load being positioned at or below the top rim of the sidewall.

[0024] Additional embodiments and/or features are set forth in part in the description that follows, and will become apparent to those skilled in the art upon examination of the specification or may be learned by the practice of the disclosed subject matter. A further understanding of the nature and advantages of the present disclosure may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure. One of skill in the art will understand that each of the various aspects and features of the disclosure may advantageously be used separately in some instances, or in combination with other aspects and features of the disclosure in other instances.

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## Description

### DETAILED DESCRIPTION OF DRAWINGS

[0025] FIG. 1A shows a front, right perspective view of a truck with an example of the mounting system attached.

[0026] FIG. 1B shows a front, right perspective view of a truck having a topper with an example of the mounting system attached.

[0027] FIG. 2 shows a rear elevation view of a truck with an example of the mounting system attached.

[0028] FIG. 3 shows a perspective view of an example of the mounting assembly.

[0029] FIG. 4A shows a side elevation view of an example of the mounting assembly secured to the flange of the sidewall.

[0030] FIG. 4B shows a side elevation view of the FIG. 4A example further extended.

[0031] FIG. 5 shows an exploded perspective view of an example of the mounting assembly.

[0032] FIG. 6 shows a cross sectional view of the interior of an example of the load support structure.

[0033] FIG. 7A shows a side cross sectional view of an example of the mounting system with the utility portion of remove for clarity.

[0034] FIG. 7B shows the coupling fasteners in a loosened position.

[0035] FIG. 7C shows the coupling fasteners in a closed position.

[0036] FIG. 8 shows an example of the mounting system with an adjustable joint.

[0037] FIG. 9 shows a side cross sectional view of an example of the adjustable joint mounting system in two positions.

[0038] FIG. 10 shows an example of the mounting assembly with no adjustment mechanism.

[0039] FIG. 11 shows an example of the mounting assembly in two positions.

[0040] FIG. **12** shows a perspective cross sectional view of an example of the mounting assembly.  
[0041] FIG. **13** shows a cross sectional view of an example of the mounting assembly.  
[0042] FIG. **14** shows an example of the mounting assembly in a further extended view.  
[0043] FIG. **15** shows an example of the load support structure.  
[0044] FIG. **16** shows an example of a spanning load support structure.  
[0045] FIG. **17** shows an example of the mounting assembly with brace members.  
[0046] FIG. **18** shows an example of the mounting assembly with a reinforcement member.  
[0047] FIG. **19** shows an example of the mounting assembly configured to directly receive a load.  
[0048] FIG. **20A** shows an example of a mounting assembly.  
[0049] FIG. **20B** shows an exploded view of the mounting assembly of FIG. **20A**.  
[0050] FIG. **20C** shows a section view of the mounting assembly of FIG. **20A** taken along section line **20C-20C** of FIG. **20A**.  
[0051] FIG. **20D** shows a section view of the mounting assembly of FIG. **20A** taken along section line **20D-20D** of FIG. **20A**.  
[0052] FIG. **20E** shows an example of the mounting assembly of FIG. **20A** in retracted configuration.  
[0053] FIG. **20F** shows the example of the mounting assembly of FIG. **20A** in a second configuration.  
[0054] FIG. **20G** shows a locking assembly for use with the mounting assemblies disclosed herein.  
[0055] FIG. **21** shows an example of a mounting assembly.  
[0056] FIG. **22A** shows an exploded view of an example of a mounting assembly.  
[0057] FIG. **22B** shows a section view of the mounting assembly of FIG. **22A**.

#### DETAILED DESCRIPTION

[0058] The present disclosure generally provides for a mounting system **100** for coupling components to the exterior of a bed **152** of a vehicle **150**. In one example, as shown in FIGS. **1A**, **1B**, **2** and **4A**, the vehicle bed **152** may in one example be a truck bed **152**. The vehicle bed may include at least one sidewall **154**, the sidewall **154** may have an inside wall and an outside wall. In general with respect to this description, an outward direction is a direction away from the interior of the truck bed **152** and an inward direction is a direction towards the interior of the truck bed **152**. The top portion of the sidewall **154** is the rim **156**, which may include a flange **158**, best shown in FIGS. **4A** and **4B**, extending downwardly from the inside edge of the rim **156**. The truck may have a truck bed cover (also referred to as a topper), such as is shown in FIG. **1B**.

[0059] The mounting system **100** includes a mounting assembly **200** attached to the vehicle, and notably to the vehicle bed, such as the sidewall **154** with a supporting structure positioned on the outside of the sidewall **154** and extending below the rim **156** so that an engagement location **226** between the mounting assembly **200** and any load **325** attached thereto is at or below the top rim **156** of the sidewall **154**. The mounting assembly **200** is attached to the sidewall **154** in such a way as to allow other components to be mounted in or on the bed **152** of the truck **150**, including on the top rim **156** of the sidewall **154**. With the supporting structure positioned on the outside of the sidewall **154**, gear or other stowage may be carried by the vehicle **150** and conveniently accessed while other objects may be positioned in the bed and/or on the top of the sidewall. Further, additional storage capacity is added to the vehicle **150** while the usefulness of the bed of the vehicle **150** is maintained. One example of the mounting system **100** includes a mounting assembly **200** and one or more load support structures **300**. Load support structures **300** include any load **335** or component **302** that may be attached or that may support, contain, store, or allow a load to be supported by the mounting assembly **200**. The load support structure **300** may be removable from the mounting assembly **200** or it may be permanently affixed to the mounting assembly **200**.

[0060] FIG. **3** shows an example of the mounting assembly **200**. The mounting assembly **200** includes a bracket **202** configured to extend outward from the sidewall **154** and to receive a load support structure **300**, a securement mechanism **253** to removably attach the mounting assembly

**200** to the sidewall **154**, as shown in FIG. 4, and an adjustment mechanism **275** to extend or retract the bracket **202** relative to the rim **156** of the sidewall **154**. In this example, the securement mechanism **253** attaches the mounting assembly **200** to the flange portion **158** of the sidewall **154**. In this example, the mounting system **100** has a reduced vertical profile and allows for additional vehicle attachments, such as for an example a truck bed cover (also referred to as a topper), to be placed on the rim **156** and over the top of the mounting system **100**, without substantially interfering with the function of the truck bed cover, such as its seal to the sidewalls **154** of the bed. In additional examples, as are further described below, the securement mechanism **253** may permanently attach the mounting assembly **200** to the sidewall **154**. In some examples, the securement mechanism **253** may releasably attach the mounting assembly to the sidewall **154**. In additional examples, as are further described below, the adjustment mechanism **275** may not be included, in which case there is no adjustment relative to the sidewall **154** of the bed. In other examples, the adjustment mechanism **275** and/or securement mechanism **253**, separately or together, are configured to reduce portions of the mounting assembly **200** that extend inwardly toward the interior of the truck bed **152**, allowing for the full volume of the truck bed **152** to be used more conveniently. For instance, this may allow the use of truck bed campers received within the truck bed **152**. Other examples may have a reduced vertical profile as well as reduced or no portions extending into the bed **152** of the truck **150**. Further benefits of the invention include at least that the user can still access the bed **152** conveniently, the load support structure **300** is positioned at a height convenient for a user to access, loads **335** may be transported that would otherwise be too large to be contained by the volume of the vehicle bed **152**, and the total storage and utility capacity of the vehicle **150** is increased.

[0061] FIGS. 3 through 7 show an example of the bracket **202**. The bracket **202** includes a top portion **204** positioned on the top of the rim **156** of the sidewall **154**, a lower portion **220** extending downward to a point below the rim **156** of the sidewall **154**, and an intersection portion **244** where the top portion **204** and the lower portion **220** intersect. In this example, the top portion **204** rests on the top of the rim **156** of the sidewall **154** and interfaces with a slider member **280**, the slider member **280** is a component of both the securement mechanism **253** and the adjustment mechanism **275**, assists with both securing the bracket **202** to the sidewall **154** of the truck bed and in positioning the bracket **202** relative to the sidewall **154**. The top portion **204**, lower portion **220**, and the intersection portion **244** may be defined as one piece, or each portion may be one or more pieces that mate to form the bracket **202**. The bracket **202** may be a thin metal plate throughout, or the bracket **202** may be assembled from one or more materials in a variety of shapes and sizes sufficient to support a load at a position at or below the rim **156** of the truck bed.

[0062] In this example, the top portion **204** defines one or more apertures **206**. In this example each aperture **206** is an elongated slot **210** defined by opposing lateral sidewalls **212** and opposing end walls. Each slot **210** is located adjacent to and may extend beyond the inside edge of the top portion **204**. A track **214** may be defined on one or both of each opposing lateral sidewall **212**. The tracks **214** are configured to receive the slider member **280** that may be positioned along the length of the slot **210** and engage the tracks **214**, as described below with respect to one example of the adjustment mechanism **275**. The track **214** portions may define a series of teeth **218** extending along at least a portion of the length of the track **214** to engage corresponding teeth **218** optionally formed on the corresponding slider, which reduces the movement of the slider in the respective slot **210**, and thus the bracket **202** position relative to the sidewall **154**.

[0063] The bracket **202** is secured to the sidewall **154** of the truck bed **152** in this example by a securement mechanism **253**. The securement mechanism **253**, as explained below, exerts a force on the slider member **280**, causing the slider member **280** to exert a force on the tracks **214** of the slots **210** and fix its position along the length of the slot **210**, and hold the bracket **202** in position relative to the sidewall **154**. If teeth **218** are present in the tracks **214**, the slider member **280** may be configured to engage the teeth **218** of the tracks **214** to further reduce movement.

[0064] The lower portion **220** of the bracket **202** connects with the top portion **204** of the bracket **202** at an intersection portion **244**. The lower portion **220** may be configured to extend at an angle Alpha ( $\alpha$ ) **246** both downward below the top rim **156** of the sidewall **154** as well as outwardly away from the sidewall **154** to form a gap **234** between the sidewall **154** and the bracket **202**. The gap **234** may help to reduce contact between the lower portion **220** and the sidewall **154** of the truck **150**. The lower portion **220** may be configured to define a receiving portion **224** defining an engagement location **226** where a load **335** or load support structure **300** is secured. The receiving portion **224** may define, in this example, one or more apertures **228** to releasably receive a corresponding coupling fastener **323** or feature on a load support structure **300**. In the example shown in FIG. 3, two apertures **228** defined by the receiving portion **224** are configured to couple the load support structure **300** to the bracket **202**. The apertures **228** may define threaded holes that correspond to coupling fasteners **323** on a load support structure **300**, further described below with respect to FIG. 5, so as to couple the load support structure **300** to the bracket **202**, and thus to the mounting assembly **200** as a whole. In other examples, the apertures **230** may not be threaded **232** or there may be a sufficient number of receiving portions **224** to couple more than one load support structure **300** to the lower portion **220**. The apertures **228** or the receiving portion **224** may also be shaped to support a load without a fastener **323**, such as being a hook **232** or clamping structure as shown by FIG. 19. Some examples may or may not use coupling fasteners **323** to secure load support structures **300** to the receiving portion **224**. In other examples, multiple receiving portions **224** may define multiple engagement locations. The multiple receiving portions **224** may include multiple aperture **228** locations or multiple features on the lower portion **220**, such as a plurality of hooks **232**, ledges or clamps.

[0065] The intersection portion **244** of the bracket **202** defines where the lower portion **220** and the top portion **204** intersect, see FIGS. 3 and 4A. In examples where the bracket **202** is one piece, the intersection may be defined by a change of direction or a bend **248** in the bracket **202**. The change in direction or bend **248** defining the intersection portion **244** may be a discrete angle, a bend **248** to reduce the concentration of forces at the bend **248**, or any other geometric relation sufficient to define a change of direction between the top portion **204** and the lower portion **220**. In examples where the bracket **202** is formed of multiple pieces, the intersection portion **244** may be the location where the pieces that define the top portion **204** interface with the pieces defining the lower portion **220**. In one example, the intersection portion **244** defines an angle Alpha ( $\alpha$ ) **246** between the top portion **204** and the lower may range from about 90 to about 180 degrees in order to form a gap **234** between the lower portion **220** and the sidewall **154** of the bed. A range of approximately 100 degrees and greater has been found to generally form a gap **234** between the lower portion **220** and the sidewall **154** and support a load **335**, though more acute or obtuse angles may also be sufficient. In a range of about **100** to **120** degrees, there is a gap **234** defined between the lower portion **220** of the bracket **202** and the sidewall **154** while keeping the load relatively close to the sidewall, and in some examples flush to or inside the lateral extension of the side mirrors on the vehicle. The gap **234** may be sufficient to reduce the risk of the bracket **202** (or bracket **202** and any load attached to the bracket **202**) deflecting into the sidewall **154** when the bracket **202** is subjected to a force, such as when the vehicle **150** passes over a pot hole on a roadway. The intersection portion **244** may further define an adjustable joint **502** to allow the orientation of the lower portion **220** to be changed relative to the top portion **204**, angle Alpha ( $\alpha$ ) **246** to allow for a wider variety of trucks and loads to be compatible with the mounting assembly **200**.

[0066] In other examples, as shown in FIG. 18, the intersection portion **244** may include a reinforced member **250** to reduce the possible deflection of the lower portion **220** relative to the top portion **204**. For example, strut members **250** may extend across the intersection portion **244** from the lower portion **220** to the top portion **204**. This reinforcing supporting structure may reduce the degree to which the bracket **202** flexes under a load during use. The brackets **202** may be made of

metal, plastic or a combination of these or other materials. Some more flexible materials may require additional reinforcement structure **250**, while other stiffer bracket **202** materials may not require reinforcement materials **250**.

[0067] The securement mechanism **253**, as shown in FIG. **3** through **7A**, secures the mounting assembly **200** to the sidewall **154**. In the example shown, the securement mechanism **253** includes a clamp member **261**, a rod **255** and a fasteners **271**. In the example shown by FIG. **3**, there are two of each component, in other examples only one of each may be used, in others still more features may be used to secure the mounting assembly **200**. This example of the securement mechanism **253** releasably couples to the flange **158** of the sidewall **154**. The rod **255**, best shown in FIGS. **4A** and **4B**, may be threaded and an upper portion **257** connects to the top portion **204** of the bracket **202**, such as by threadedly engaging an aperture **206** formed in the top portion **204** of the bracket **202**. In some examples the aperture **206** is formed in a slider member **280** positioned in a slot **210** formed in the top portion **204**, which are part of the adjustment mechanism **275** described further below. A bottom portion **258** of the rod **255** engages a clamp member **261**, such as by extending through an aperture **284** formed therein. A fastener **271** threadedly engages the bottom portion **258** of the rod **255** to cause a compressive force between the top portion **204** of the bracket **202** and the clamp member **261**, which secures the mounting assembly **200** to the sidewall **154**. The clamp member **261** may include a U-shaped component **265**, such that the clamp member **261** is configured to receive a bottom portion of the flange **158**, and a receiving block **267** that receives both the U-shaped component and the bottom portion **258** of the rod **255**. In one example the fastener **271** is a wing nut. As the fastener **271** is rotated along the threaded rod **255** it contacts the bottom of the clamp member **261**. Further rotating the fastener **271** exerts a force such that the clamp member **261** exerts a compressive force on the bottom of the flange **158** of the sidewall **154** through the clamp member **261** and on the top of the rim **156** of the sidewall **154** through the top portion **204** of the bracket **202**. These forces removably secure the mounting assembly **200** to the vehicle **150** and reduce movement of the mounting assembly **200**. In other examples, the securement mechanism **253** may couple directly to other features of the sidewall **154**, such as the top rim **156**. The securement mechanism **253** may also be permanent in other examples and/or use more or less components to secure the mounting assembly **200** to the sidewall **154**.

[0068] The adjustment mechanism **275** may allow for the bracket **202** to extend outward or inward relative to the top rim **156** of the sidewall **154**, as shown by Position A—**285** in FIG. **4A** and position 2 in FIG. **4B**. In one example, as shown by FIG. **3**, the adjustment mechanism **275** includes a slider member **280** that fits within a slot **210** defined in the top portion **204** of the bracket **202**. The slider member **280** rests on tracks **214** formed on opposing sidewalls within the slot **210**. As noted above, the slider member **280** may define an aperture **284**, the aperture **284** receiving a threaded rod **255** as part of the securement mechanism **253** described above. The slider member **280** and the corresponding components may have one of a multiple of shapes and sizes, and here are shown as generally rectangular in shape. The slider member **280** as shown by FIG. **3** is configured to fit securely in the slot **210** with little additional movement. The bottom of the slider may define one or more ridges, teeth, or other structures **282**, best shown in FIG. **3**, on opposing edges **212** to interface with features **282** on the opposing tracks **214** in the slot **210**. For instance, the position of the slider member **280** along the length of the slot **210** may be set at discrete locations, **285** and **286**, to reduce the likelihood of unintended movement of the slider within the slot **210**. The engagement of the features **282** with the tracks **214** may also enhance the securement mechanism **253**. When the securement mechanism **253** is loosened, the slider member **280** may be adjusted from a first position **285**, such as that shown in FIG. **4A**, to a second position **286** shown in FIG. **4B**. In alternative examples the tracks **214** and slider member **280** may both be smooth and rely on friction to prevent movement. In additional examples, the adjust mechanism **275** may also allow for adjustment in one or more directions in addition to adjustments that position the bracket **202** more outwardly or inwardly. Other examples of the adjust mechanism allow for the bracket



**202** to move relative to the sidewall **154** with reduced interference with the volume of the truck bed **152**. The adjustment mechanism **275** is an optional feature of the mounting assembly **200**. [0069] FIGS. 5-7 show an example of the load support structure **300** that releasably attaches to the mounting assembly **200** in order to allow articles to be carried outside of the truck bed **152**. The load support structure **300** may include a utility portion **302** and a coupling portion **312**. The utility portion **302** may be a structure that supports, encloses, or otherwise allows the mounting assembly **200** to support a load **335** on the exterior side of the truck bed **152**. In the example shown in FIG. 5, the utility portion **302** is a container. The container may have a lid and a closure mechanism that allows access to an interior storage compartment **310**. The interior storage compartment **310** may be used to store a variety of articles. In other examples, the utility portion **302** may be a hook **312**, a platform, an eyelet **312**, or a combination of structures that may be used to store or support a load. The load support structure **300** may include a coupling portion **312** that engages the receiving portion **224** of the mounting assembly **200**. The engagement location **226** of the mounting assembly **200** is positioned at the same level as the rim **156**, or at a level below the rim **156**, of the sidewall **154**, such that the coupling portion **312** engages the mounting assembly **200** at the same level as the rim **156**, or at a level below the rim **156**, of the sidewall **154**. This position of engagement has many benefits, including at least that the user can access the bed conveniently, the utility portion **302** is positioned at a height convenient for a user to access, loads may be transported that would otherwise be too large to be contained by the volume of the vehicle bed, and the total storage and utility capacity of the vehicle **150** is increased. In general, and with respect to the examples described throughout, a location of the coupling portion **312** below the level of the rim **156** may in some cases be especially beneficial because that location may leave the rim **156**, and access to the bed or the attachment of other structures to the truck, over or near the rim **156**, more unobstructed or more likely unobstructed.

[0070] The coupling portion **312** in the instant example is shown in FIGS. 5-7, and includes a mounting plate **317** which is attached to the utility portion, and the mounting plate **317** defines one or more apertures **319**, where each corresponds to at least one of the apertures **228** defined on the receiving portion **224** of the bracket **202** of the mounting assembly **200**. The mounting plate **317** may have a substantially planar face **315**. The face **315** may be suitable to receive a coupling structure. For example, a coupling fastener **323** may be received in each of the plate apertures **319**. The load support structure **300** in this example is a container, such as a bag. The mounting plate **317** is attached to, or configured integrally with, a wall of the container that is intended to be positioned closest to the sidewall **154**. In this example the coupling fasteners **323** may be located on the interior of the bag, see FIG. 6.

[0071] As shown in the example of FIG. 7, to attach the load support structure **300** to the mounting assembly **200**, the container is positioned adjacent the mounting assembly **200**, and specifically the mounting plate **317** is aligned with the receiving portion **224** of the lower portion **220** so that plate apertures **319** are aligned with the corresponding apertures **228** in the receiving portion **224** of the lower portion **220**. A coupling fastener **323** is received in each aligned plate aperture **319** and aligned receiving aperture **228**, and secures the mounting plate **317** to the receiving portion **224** of the lower portion **220**. In this example, the coupling fastener **323** is a threaded cam locking bolt **323** which is received through the plate aperture **319**, and threadedly engages with the threaded receiving aperture **319** on the lower portion **220** of the bracket **202**. A suitable threaded cam locking bolt **323** may be McMaster-Carr Clamping Handle with Threaded Stud, part number 5720K25. When the threaded cam locking bolt **323** is sufficiently threadedly engaged, the lever **325** on the outer end of the cam locking bolts **323** may be actuated **326** to clamp draw the receiving portion **224** of the bracket **202** towards the outer end of the cam locking bolt **323** and clamp **327** the mounting plate **317** of the container between the outer end of the cam bolt **323** and the receiving portion **224** of the bracket **202**. This secures the load support structure **300** to the mounting assembly **200**. In other examples, the load support structure **300** may not have a mounting plate

**317.** In yet other examples, the load support structure **300** may not include a coupling structure corresponding to the mounting assembly **200** receiving portion **224**. In those examples, the mounting assembly **200** may have features that interface with the load support structure **300** directly. In other examples of the mounting system **100**, no separate load support structure **300** may be used, instead the mounting assembly **200** may directly support a load, as shown by FIG. **19**. In these examples, the load support structure **300** may attach directly to the receiving portion **224** of the mounting assembly **200** by way of fastening members, such as straps, bungee cords, zip ties, or other such structures. For example, where the load support structure **300** is an article, such as a shovel or other item or items, the shovel may be attached to the mounting assembly **200** by fastening members.

[0072] Further, multiple mounting systems **100** may be used in conjunction to support one or more loads. The load support structure **300** may be adjusted to correspond to a spanning load **330**, or the load support structure **300** itself may be designed to span between two or more receiving portions **224** or to support a spanning load **330** between two or more load support structures, as shown in FIG. **17**. In one example, the mounting system **100** may include a first mounting assembly **200** towards the front of the bed and a second mounting assembly **200** towards the rear of the bed and a single load support structure **330** with at least two coupling portions **312**, the coupling portions **312** corresponding to the first and second mounting assemblies, and the load support structure **300** further secured to both the first and second mounting assemblies. By using more than one mounting assembly **200** in a mounting system **100**, heavier or larger loads **330** may be supported than would otherwise be possible with a single mounting assembly **200**.

[0073] In other examples of the mounting system **100**, the lower portion **220** of the bracket **202** may optionally include a brace member **236**, as shown in FIG. **17**, extending between the lower portion **220** of the bracket **202** and the sidewall **154**. The brace member **236** may support the load support structure **300** against the sidewall **154** of the vehicle **150** and may also be a coating or pad **240** to reduce wear on the sidewall **154** of the truck **150**. The brace member may be a discrete strut, a peg, or some other structure **238** that extends from the lower portion **220** of the bracket **202** in the inward direction and may engage, continuously or only upon deflection of the lower portion **220** of the bracket **202**, the outward portion of the truck sidewall **154**. The brace member **236** may extend along all or a portion of the length of the inward side of the lower portion **220**, where an engagement portion of the brace feature **238** may contact the sidewall **154** of the truck **150**. In such an example, the brace member **236** may be an impact resistant foam **240**. Where the bracket **202** includes a brace member **236**, the brace member **236** may allow for the bracket **202** to support a heavier or larger load support structure **300** than would otherwise be possible if the brace member **236** was not present. Additionally, or separately, the lower portion **220** of the bracket **202**, or the brace member **236**, may have a coating **240**, such as a padded or resilient materials such as rubber or plastic **240**. The coating **240** may be positioned on the lower portion **220** of the bracket **202**, or on the brace member **236**, at a location where each may contact the sidewall **154** of the truck. In some examples, the size or position of the brace member **236** may be adjusted so as to form a gap **234** between the lower portion **220** and the sidewall **154** of the truck when in the unloaded or nominally loaded position.

[0074] In some examples of the mounting system **100**, as shown in FIGS. **8** and **9**, the intersection portion **244** of the bracket **202** is adjustable **500** and may allow for the angle between the top portion **204** and the lower portion **220** of the bracket **202** to be adjusted. The angle Alpha ( $\alpha$ ) **246** of the bracket **202**, defined as the angle between the top portion **204** of the bracket **202** and the lower portion **220** of the bracket **202**, may be adjusted to many angles between contacting the sidewall **154** of the truck and a location under 180 degrees, where 180 degrees would correspond to the lower portion **220** and the top portion **204** of the bracket **202** forming a straight line. Other examples may include one or more variations of the components elsewhere described in this specification in addition to the examples currently described.

[0075] In one example, the adjustable intersection portion **500** may be a joint **502**, such as in one example a piano hinge, where the top portion **204** of the bracket **202** is configured to define one or more knuckles **504** of the hinge and the lower portion **220** of the bracket **202** is configured to define one or more knuckles **506** of the hinge, the knuckles **504** of the top portion **204** configured to alternately correspond to the knuckles **506** of the lower portion **220**. The joint **502** may further include a cam-pivot axle **510**, which may include an axle **510** defining threading at one end and a cam locking lever **514** at the opposing end, and a collar **512** configured to slide over the axle **510** to be placed on the locking lever **514** side of the axle **510**. When the adjustable intersection portion **500** is assembled, the knuckles of the top **504** and lower **506** portion **220** interlineate to allow the axle **510** of the cam-pivot mechanism **500** to pass through the knuckles, **504** and **506**. The end knuckle **504** and **506**, may be threaded to threadedly receive the threaded end of the axle **510**, or there may be an additional fastener, such as a nut, configured to thread over an exposed portion of the axle **510** that may extend outward from the end knuckle, **504** and **506**. The adjustable intersection **500** may be positioned from an original location **516** and set at desired angle **517** Alpha ( $\alpha$ ) **246**, and then secured at that angle **517** by actuating the cam locking lever **514**. When the cam lock lever **514** is engaged, the knuckles, **504** and **506**, of the joint **502** are compressed together such that movement of the lower portion **220** of the bracket **202** relative to the top portion **204** of the bracket **202** is inhibited.

[0076] Other examples of the adjustable intersection portion **500** may allow for pre-configured bottom or top portion **204s** to be replaced to change the angle Alpha ( $\alpha$ ) **246** or the general configuration of the intersection portion **500**.

[0077] In another example of the mounting assembly **200**, as shown in FIGS. **11** to **14**, the mounting assembly **1000** may include a bracket housing **1200** including a securement mechanism **253** and an adjustment mechanism **1300**. In this example the engagement location **1400** of the load support mechanism, as with the other example above, is at or below the level of the truck bed rim **156**. Further, the coupling mechanism may be the same as an earlier example, or, additional or different components may be further included. This example may include a brace member **236** as previously described in this specification, and the components of the bracket **1100** assembly may further include a coating as previously described. In this example, the mounting assembly **1000** is a receiving housing assembly **1000** and includes a multi-layered bracket housing **1200** and a bracket **1100** having a top portion **1102** and a lower portion **1106**, similar to the bracket **1100** described with respect to FIGS. **1-7**. The configuration of this example of the mounting assembly **1000** allows for the bracket **1100** to extend and retract inward and outward relative to the rim **156** of the sidewall **154** without a portion of the bracket **1100** extending into the volume of the truck bed **152**. The reduced interference with the volume of the truck bed **152** allows for objects that fit into the truck bed **152** and are adjacent to the inside edge of the sidewall **154**, such as truck bed campers, utility boxes, and other payloads, to be used in conjunction with the mounting system **100**. Other examples may include one or more variations of the components elsewhere described in this specification in addition to the examples currently described.

[0078] The bracket housing **1200** includes a top plate **1202**, a bottom plate **1220**, and an elongated recess **1250** defined between the top plate **1202** and the bottom plate **1220**. In this example, the bottom plate **1220** is planar, and the top plate **1202** has one or more opposing edge portions **1204** that generally align with one or more opposing edges **1224** of the bottom plate **1220**. One or more central portions **1210** of the top plate **1202**, between the opposing edge portions **1204**, is displaced upwardly from the plane of the opposing edges **1224**. A gap **234** between a central portion **1210** of the top plate **1202** and a central portion **1225** of the bottom plate **1220** forms an elongated recess **1250**. The elongated recess **1250** is configured to receive the top portion **1102** of the bracket **1100**.

[0079] This example of the mounting assembly **1000** is coupled directly to the top rim **156** of the sidewall **154**. The bracket housing **1200** may be permanently or removably coupled. As shown by FIG. **12**, the aligned outer edges of the top plate **1202** and the bottom plate **1220** define securement

portions **1204**, each of which include one or more securement apertures **1206** and **1226**. One or more of the securement apertures **1206** and **1226** may correspond to openings on the top of the rim **156** of the sidewall **154**. A securement fastener **1240**, such as a threaded fastener, may be inserted into the corresponding securement apertures **1206** and **1226** and openings in the rim **156** of the sidewall **154** and couple the mounting assembly **1000** to the rim **156** of the sidewall **154**. The securement apertures **1206** and **1226** and the openings may be threaded or an additional fastener, such as a nut, may receive the securement fastener **1240** to attach the mounting assembly **1000** to the rim **156** of the sidewall **154**.

[0080] The adjustment mechanism **1300** may secure the top portion **1102** of the bracket **1100** inside the elongated recess **1250**, and allow for adjustment of the amount the top portion **1102** is received within the recess. The positioning of the top portion **1102** of the bracket **1100** within the recess defines the extension of the bracket **1100** relative to the sidewall **154** of the truck bed.

[0081] In this example, the bracket **1100** portion is similar to previously described examples of the bracket **202**, and includes a top portion **1102**, lower portion **1106**, and an intersection portion **1104**. The top portion **1102** may also define one or more adjacent apertures. In the example shown in FIGS. **11** to **14**, the top portion **1102** defines one or more adjustment apertures **1305**. One or more apertures, **1306** and **1307**, are formed in the top and bottom plate **1220s**, and correspond to the location of the adjustment apertures **1305** in the top portion **1102** of the bracket **1100**. In this example, the adjustment apertures, **1305**, **1306**, and **1307**, are aligned linearly **1310** with regular spacing there between. Many other aperture patterns **1310** would suffice so long as the pattern **1310** allows for the bracket **1100** to be repositioned inwardly **1355** and outwardly **1350** relative to the rim **156** of the sidewall **154**, and be secured by corresponding aligned apertures **1306** and **1307** in the top **1202** and bottom plates **1220**. The adjustment apertures **1305** in the top portion **1102** of the bracket **1100** may be threaded or define un-threaded holes

[0082] As shown by FIGS. **13** and **14**, the adjustment apertures **1307** on the bottom plate **1220**, most visible in FIGS. **13** and **14**, may be threaded while the apertures **1306** on the top plate **1202** may be countersunk and threaded or unthreaded. In the configuration described, a threaded adjustment fastener **1308** may be inserted into the corresponding adjustment apertures, **1305**, **1306**, and **1307**, to hold the bracket **1100** in a set location relative to the bracket housing **1200** and the sidewall **154**. The mounting assembly **1000** of this example maintains a low profile designed to reduce interference with items engaging the top rim **156** of the sidewall **154**.

[0083] The apertures **1306** and **1307** of the multi-layered housing may be configured to allow for the bracket **1100** to be adjusted into positions further inward **1355** or outward **1350** relative to the sidewall **154**, as shown by FIG. **11**. Adjustment of the bracket **1100** may be accomplished by moving the bracket **1100** further into or extend from the recess, aligning at least one adjustment aperture **1305** in the bracket **1100** with at least one corresponding aperture **1306** in the top plate **1202**, and securing the bracket **1100** in place with at least one adjustment fasteners **1308**. In one example, the adjustment fastener **1308** may extend only into the top portion **1102** of the bracket **1100**. In other examples, the adjustment fastener **1308** may extend through the top plate **1202** and into the bottom plate **1220**, through the top portion **1102** of the bracket **1100**. When the mounting assembly **1000** is in the fully retracted corresponding position, shown by FIG. **13**, the top portion **1102** of the bracket **1100** may not extend into the interior volume of the truck bed **152** and interfere with use of the truck bed **152**.

[0084] In the example shown in FIG. **10**, the mounting assembly **200** includes a bracket **600** and one or more fasteners **650**. In this example the bracket **600** includes a top portion **602**, a lower portion **620**, an intersection portion **630**, and may include a securement portion **610**. In this example the bracket **600** is attached directly to the sidewall **154** by fasteners **650**. The distance of the lower portion **620** of the bracket **600** to the sidewall **154** may not be adjustable. The angle of the lower portion **620** of the bracket **600** may or may not be adjustable relative to the top portion **602** similar to the example in FIGS. **8** and **9**. Other examples may include one or more variations of

the components elsewhere described in this specification in addition to the examples currently described.

[0085] The top portion **602** of the bracket **600** rests on the rim **156** of the sidewall **154** and may define apertures **605**. The lower portion **620** may be similar to previous examples **220** and may include a receiving portion **622**, the receiving portion **622** defining an engagement location **624** between a load **335** or a load support structure **300**. The engagement location **624** is at or below the rim **156** of the sidewall **154**. The receiving portion **622** may define one or more apertures **626**. The intersection portion **630** may also be similar to previous examples, defining the location where the top portion **602** and the lower portion **620** intersect, as well as the angle Alpha ( $\alpha$ ) **246** between the top portion **602** and the lower portion **620**. A gap **234** may be defined by the area between the sidewall **154** and the lower portion **620** of the bracket **600**. The intersection may be continuous or may be defined as the location where one or more separate portions defining the top and lower portions **620** interface. The securement portion **610** may be a portion that extends from or interfaces with the top portion **602** of the bracket **600** extending downward relative to the rim **156** of the truck bed along the flange **158** of the sidewall **154**, with a portion **610** extending outward under the flange **158**. One or more securement apertures **605** may be defined by the securement portion **610** such that a fastener may be inserted to secure the bracket **600** to the flange **158**.

[0086] This example of the mounting assembly **200** includes a reduced vertical profile and a reduced horizontal profile, allowing objects such as truck covers or truck bed campers to be placed on the rim **156** and over the top of the mounting system **100**, and sealed, as well as against the mounting system **100** in the truck bed **152** with reduced interference. In other examples, there may be no securement portion **610** and the securement mechanism **253** may be defined by securement fasteners **650** inserted into apertures **605** on the top portion **602** to secure the top portion **602** of the bracket **600** to the rim **156** of the truck bed. In this example, there would be little or no interference with the volume of the truck bed **152** while the reduced vertical profile would allow for objects to still be placed over the top of the mounting system **100** and on the rim **156** of the sidewall **154**.

[0087] In other examples of the load support structure **300**, the utility portion **302** may contain one or more useful features, such as a hook **312**, or an eyelet **312** as shown in FIG. **15**. The load support structure **300** may be configured to allow for multiple utility portions to be connected to the same coupling portion **312**.

[0088] Regarding each of the examples described herein, while the engagement location **226** between the coupling portion **312** of the load support structure and the receiving portion **224** of the mounting assembly **200** is at or below the rim **156** of the sidewall **154**, the load carried by a load support structure **300** may extend above the rim **156** of the sidewall **154**, as shown by FIG. **16**.

[0089] Other examples of a mounting assembly **2000**, **2100**, and **2200** are shown for example in FIGS. **20A-20G**, **21**, **22A**, and **22B**, respectively. The mounting assembly **2000**, **2100**, **2200** may be similar to other mounting assemblies disclosed herein and have similar benefits. These benefits include conveniently positioning cargo and accessories **2008** outboard of a vehicle **150** body, and alternatively or additionally, outside of a cover component mounted on a vehicle body, such as for example a topper structure (see FIG. **1B** and FIG. **20C**).

[0090] As shown for example in FIG. **20A**, and as may apply also to FIGS. **21**, **22A** and **22B**, the mounting assembly **2000** may be mounted to a rim **156** of a sidewall **154** of a truck bed **152**. The mounting assembly **2000** allows the attachment of a load support structure **2242** (for example, a container as shown) to the outside of a vehicle for easy access. The coupling portion **2240** of the load support structure **2242** may be releasably attached with the receiving portion **2244** of the mounting assembly **2200**. Because the receiving portion **2244** defines the engagement location **2238** of the mounting assembly **2000**, and the receiving portion **2244** is in one example below the rim **156**, or in another example is at the level of the rim or below the rim **156**, the engagement location **2238** is at or below the rim **156** of the sidewall **154** consistent with the other examples of the mounting assembly described herein. The cargo attached to the load support structure **2242** may

be easily accessible from outside the truck bed.

[0091] Continuing with FIG. **20A**, and as may also apply to FIGS. **21**, **22A** and **22B**, the load support structure **2242** may include a utility portion **2250** and a coupling portion **2240**. The utility portion **2250** may be a structure that supports, encloses, or otherwise allows the mounting assembly **200** to support a load **335** on the exterior side of the truck bed **152**. In the example shown in FIG. **20A**, the utility portion **2250** is a container. The container may have a lid and a closure mechanism that allows access to an interior storage compartment. The interior storage compartment may be used to store a variety of articles. In other examples, the utility portion **2250** may be a hook (see FIGS. **15**, **19**), a platform, an eyelet (see FIG. **15**), or a combination of structures that may be used to store or support a load. The load support structure **2242** may include a coupling portion **2240** that engages the receiving portion **2244** of the mounting assembly **2000**. The engagement location **2238** of the mounting assembly **2000** is positioned at the same level as the rim **156**, or at a level below the rim **156**, of the sidewall **154**, such that the coupling portion **2240** may engage the mounting assembly **2000**, **2100**, **2200** at least partially at the same level as the rim **156**, or at a level below the rim **156**, of the sidewall **154**. The coupling portion **2240** may be integrated with the load support structure **2242**, or may be separate from the load support structure **2242** and removably attached there to. The coupling portion may be secured to the receiving portion **2244** by fasteners **2020** positioned through apertures **2021**, or by other capable attachment mechanisms.

[0092] In this example, as best shown in FIGS. **20A**, **20B**, **20C**, the mounting assembly **2000** includes a carriage assembly **2034** that is adjustable relative to the truck bed to which it is attached. The carriage assembly **2034** includes a bracket **2002** (FIGS. **20B**, **20C**) on which a carriage **2004** is movably coupled. The bracket **2002** may removably secure the mounting assembly **2000** to the vehicle **150**. The carriage assembly **2034** may also include lower portion **2006** adjustably mounted to the carriage **2004**, which in turn couples with a load support structure **2242**, and in one example is positioned outwardly from the sidewall **154** of the vehicle **150**. In one example, the lower portion **2006** may be selectively couplable to coupling portion **2240**, such as adapter **2010**. The coupling portion **2240** may be separate from or may be integrated into a load support structure **2242** (such as to facilitate the attachment of different types of cargo on the outside of the vehicle body). An adapter **2010** may be useful in that it may facilitate a coupling portion **2240** to attach to a receiving portion **2244** in the event the coupling portion **2240** and receiving portion **2244** may not easily attach together without the adapter **2010**.

[0093] As best shown in FIG. **20B**, the bracket **2002** may be formed of an upper portion **2040**, a receiving block **2044** and/or a cleat **2046**. The upper portion **2040** may be substantially planar. A clamp **2048** may operably engage with the receiving block **2044** and cleat **2046** to releasably secure the bracket **2002** to the rim **156** of the truck bed **152** without requiring the formation of apertures in the vehicle body, as described below. This is referred to herein as a clamp-bracket structure. Alternately or in combination with the bracket clamp **2048** engagement, a fastener or fasteners may extend through the bracket **2002** and through apertures formed in the rim **156** to secure the bracket to the rim **156** of the truck bed **152**.

[0094] The use of the clamp **2048**, along with the cleat **2046** to secure the bracket **2002** to the truck bed is described with respect to FIGS. **20B** and **20C**. The receiving block **2044** on the bracket **2002** may extend downwardly from the upper portion **2040**. The receiving block **2044** may have a plate-shape, and be as wide, less wide, or more wide than a width of the upper portion. The receiving block **2044** may be integrally formed with or securely attached to the upper portion. The receiving block **2044** may extend downwardly from the upper portion **2040** in a position closely adjacent to, either in contact with or spaced apart from an inside edge of the rim **156** of the sidewall **154** of the pickup truck bed **152**. The receiving block **2044** may extend at an angle toward or away from the sidewall of the truck bed **152**, based on the physical design of the rim **156** of the truck bed sidewall **154**. The angle may range from about 45 degrees towards the sidewall, to about 45 degrees away from the side wall, and may be an angle of about 90 degrees. A cleat **2046** may be coupled to or

integrally formed with the receiving block **2044**, and may be spaced downwardly from the upper portion **2040** a distance larger than the depth of the rim **156**. In some examples the cleat **2046** may be spaced about 2 inches to about 4 inches below the upper portion **2040**. The cleat **2046** may be positioned at or near the end of the receiving block **2044**. The cleat **2046** may extend at a normal angle with respect to the receiving block **2044**, or may be angled upwardly towards the rim **156**, or may be angled downwardly away from the rim. In some embodiments, the upper portion **2040** and the cleat **2046** may be substantially parallel with respect to one another.

[0095] Referring still to FIGS. **20B** and **20C**, the mounting assembly **2000** may include an inner clamp body **2048** configured to couple to the bracket **2002**. The inner clamp body **2048** may have a flange **2054** and a shoulder **2056** coupled to one another by a web portion **2066**. The flange **2054**, shoulder **2056**, and web portion **2066** may form a channel **2052** in the inner clamp body **2048**. A lip **2058** may extend from the shoulder **2056**, and may extend away from the web portion **2066** further than the flange **2054**. One or more apertures **2050** may be formed in the shoulder **2056** suitable to mount the inner clamp body **2048** to the receiving block **2044**. Referring to FIG. **20C**, the rim **156** may be securely clamped between the inner clamp body **2048** and the receiving block **2044** when the inner clamp body **2048** is coupled to the bracket **2002** by one or more fasteners **2064**. For example, the clamp body **2048** may be positioned with the lip **2058** contacts the receiving block **2044**, such as in a transition between the cleat **2046** and the receiving block **2044**, and the flange **2054** engaging an inside feature of the rim **156**. When the fasteners **2064** are tightened to draw the clamp body **2048** towards the receiving block **2044**, the clamp body levers around the contact of the lip **2058** with the receiving block **2044** to create a compressive force at the contact point between the flange **2054** and the inside of the rim **156**. A benefit of the inner clamp body **2048** and bracket **2002** may be the ability to mount the mounting assembly **2000** to the vehicle in a removable manner that does not alter the vehicle such as by drilling holes, etc. The channel **2052** may provide a clearance between the rim and the inner clamp body **2048** to accommodate a variety of vehicle types.

[0096] Referring to FIGS. **20B**, alternatively or additionally to the clamp-bracket structure, one or more apertures **2042** may be formed in the upper portion **2040** for use with fasteners **2088** to attach the bracket **2002** to the rim **156**. For example, the apertures **2042** may be laterally spaced apart in a middle portion of the upper portion **2040**. In the example in FIG. **20D**, fasteners **2088**, such as threaded bolts, may be positioned through the apertures **2042** and through corresponding apertures in the rim of the pickup bed, and secured therein by corresponding nuts. The apertures **2042** may be countersunk, such that fasteners **2088** received therein are flush with a top surface of the upper portion **2040** so as to not interfere with the slidable movement of the carriage **2004** on the bracket **2002**.

[0097] Referring to FIGS. **20A**, **20C**, and **20E**, carriage **2004** is selectively movably supported upon the bracket **2002**. For example the carriage may move outwardly away from the sidewall **154** of the truck bed **152** or inwardly toward the sidewall **154** and the truck bed **152**, in order to adjust the location of the lower portion **2006** relative to the sidewall **154**. In the example shown, there are two groups of apertures **2018** formed in the carriage **2034**. There is one group near each end of the carriage **2034**. The apertures **2018** in each group are equally spaced apart and linearly aligned within each group, and the two groups of apertures are linearly aligned with one another. Apertures **2019** are formed in the bracket **2002**, with one aperture **2019** positioned adjacent either end of the bracket **2002**. The apertures **2019** are spaced apart a distance so that each aperture **2019** aligns with one aperture the group of apertures in the respective end of the carriage **2004**. With an aperture **2019** aligned with an aperture **2018** in the carriage **2034**, a fastener **2016** (such as a bolt and associated nut) may be positioned through the aligned apertures to removably secure the carriage **2004** to the bracket. With three apertures **2018** in each of the groups of apertures, this provides three relative positions of the carriage **2004** relative to the bracket **2002**. In another example, one grouping of apertures **2018** in the carriage **2004** that aligns with an aperture **2019** in the bracket

may be replaced by a pin-and-slot structure, requiring the user to loosen only one fastener to adjust the relative positioning of the carriage **2004** with the bracket **2002**. More or fewer apertures in the carriage **2004** and/or the bracket may optionally be used, as long as one aperture **2018** in the carriage **2004** aligns with one aperture **2019** in the bracket **2002**. Alternatively, other adjustable attachment configurations may be utilized to control the adjustability of the carriage **2004** with the bracket **2002**, for example a clamp structure to pinch the carriage **2004** together with the bracket **2002**. Alternatively, the carriage **2004** may have only one position with respect to the bracket **2002**. [0098] FIG. **20C** shows the carriage **2004** positioned in the outwardmost position of the carriage **2004** relative to the bracket **2002** given the patterns of apertures **2018** in the carriage **2004**. One fastener **2018** is positioned in through the inner-most aperture **2018** of each grouping of three apertures. This positions the lower portion **2006** of the carriage assembly **2034** furthest away from the wall **154** of the truck bed **152**. FIG. **20E** shows the carriage **2004** positioned in the innermost position of the carriage **2004** relative to the bracket **2002** given the patterns of apertures **2018** in the carriage **2004**. One fastener **2018** is positioned in through the outer-most aperture **2018** of each grouping of three apertures. This positions the lower portion **2006** of the carriage assembly **2034** closest to the wall **154** of the truck bed **152**.

[0099] Continuing with reference to FIGS. **20C** and **20E**, a sidewall **2090** of a topper **2092** mounted on a truck bed **152** is shown. The sidewall **2090** is aligned generally above the sidewall **154** of the bed **154**, and is shown as being the same or similar width as the sidewall **154**, but may be wider or narrower depending on the design of the topper and the particular truck. The bracket **2002** may be designed such that the ends of the bracket extend beyond opposing walls of the sidewall **154** of the truck bed **152** and the sidewall **2090** of the topper **2092**. The aperture **2019** adjacent each end of the bracket **2002** may also be exposed on either side of the topper **2092**, such as for example being spaced away from the opposing sidewall **152** of the truck bed **154** and the sidewall **2090** of the topper **2092**, as shown in FIGS. **20C** and **20E**. This allows the fasteners in the apertures **2019** to be accessed to allow adjustment of the carriage **2004** relative to the bracket **2002** inwardly or outwardly relative to the sidewall **154** of the truck bed **152** even with a topper positioned on the truck bed **152**, and re-secured in the new position. This allows the mounting assembly **2000** to be adjustable outwardly and inwardly (for example outward away from the sidewall **154**, or inward toward the sidewall **154**) relative to the truck bed without removing the mounting assembly from the truck bed **152**, and/or without removing the topper **2092** from the truck bed **152**. The carriage assembly **2034** is also thusly adjustable outwardly and inwardly relative to the sidewall **154** without removing the carriage assembly from the truck bed **152**. As shown in FIG. **20E**, the lower portion **2006** may be pivoted outwardly about pivot formed between the carriage and the lower portion **2006**, and secured in that angled orientation by the adjustment mechanism as described further below.

[0100] Where a topper is used on a truck bed **152** that also implements one or more mounting assemblies as described herein, a spacer strip may be positioned on the rim **156** between the one or more mounting assemblies to aid in sealing the topper to the rim **156**.

[0101] One example of adjustably coupling the carriage **2004** with the bracket **2002** is shown in FIGS. **20B** and **20D**. The upper portion **2040** of the bracket **2002** may define one or more rails along opposing edges, such as rail **2060a** and rail **2060b**. The rails are each spaced above the rim **156** upon which the bracket **2002** is attached. The carriage **2004** may define one or more flanges along opposing edges, such as flange **2068a** and flange **2068b** that together form a track **2070**. The flanges may be spaced away from a lower face **2082** of the carriage **2004** and extend inwardly, forming a gap **2084** between each flange and the carriage **2004**. The rails **2060a** and **2060b** fit in the gap **2084** above the respective rails **2068a**, **2068b**, and the carriage **2004** may move relative to the bracket **2002**, and be laterally held in position by the track **2070**.

[0102] Benefits of a carriage assembly **2034** including a carriage **2004** movably mounted on the bracket **2002** may include the ability to mount differently sized cargo to a variety of different



vehicle **150** shapes and sizes. For example, larger or heavier cargo can be mounted further from the vehicle body such that the cargo is less likely to contact the vehicle body as the vehicle drives. Alternatively or additionally, a benefit of the carriage assembly **2034** may be the ability to adjust the mounting assembly **2000** without removing the carriage assembly **2034** from the truck bed **152**, or without disassembling the carriage assembly **2034**, or without removing a bed topper **2092** from the truck bed **152**, should one be installed over the mounting assembly **2000**.

[0103] As shown in FIG. **20A**, the lower portion **2006** of the mounting assembly **2000** may be pivotally coupled to the carriage **2004** via a pivot **2012**. The pivot **2012** may have the benefit of enabling the lower portion **2006** to be positionable in different angles ( $\alpha$ , shown e.g., in FIG. **20C**) with respect to the carriage **2004** to accommodate differently sized cargo, accessories, or vehicle types/sizes.

[0104] With reference to FIGS. **20A** and **20B**, the carriage **2004** may have a body portion **2036** and a head portion **2032** disposed at one end portion of the body portion **2036**. The body portion **2036** may define a substantially planar body, plate, or structure. The body portion **2036** may have formed therein one or more apertures **2018**, as referenced above. In one example, the head portion **2032** may have a body **2033** that extends across the lateral width of the body portion **2036**, and that may extend away from both opposing surfaces of the carriage **2034**, forming a general T-shape in combination with the body portion **2036**. In one example as shown in FIG. **20C** the head portion **2032** protrudes above a top surface of the carriage, and protrudes below the opposite bottom surface of the carriage **2034**, defining a long dimension of the head portion. The head portion **2032** may have one or more pivot apertures **2038a** and **2038b** (one at each end of the head portion) and/or one or more adjustment apertures **2014a** formed therein. The pivot aperture **2038a** may be positioned adjacent a top of the head portion **2032**. In one example the pivot aperture **2038a** is in the upwardly protruding portion of the head portion **2032**. The adjustment aperture **2014a** may be positioned adjacent a bottom of the head portion **2032**. In one example the adjustment aperture **2014a** is in the downwardly protruding portion of the head portion **2032**. The pivot aperture **2038a** and adjustment aperture **2014a** may extend at least partially into or all the way through the head portion **2032**, for example in a lateral or width direction relative to the length of the carriage **2004**.

[0105] Referring to FIGS. **20A**, **B**, and **F**, the lower portion **2006** may have a U-shaped cross section configured to receive the head portion **2032** of the carriage **2004**. For example, the lower portion **2006** may be formed of a transverse web **2078** that extends between two legs or flanges **2076** that extend away from the web **2078** at an angle, such as for example  $90^\circ$ . The legs **2076** may have one or more apertures formed therein. For example, the legs **2076** may have a pivot aperture **2038b**, one or more adjustment aperture **2014b**, and/or a security aperture **2022** formed therein. The legs **2076** and apertures formed therein may be substantially mirrored between the two legs **2076**. The web **2078** may have one or more apertures **2021** that form the receiving portion **2244** of the mounting assembly **2000**, which connects with the coupling portion **2240** of a load support structure **2242**. The receiving portion **2244** may define the engagement location **2238** as being at or below the rim **156** of the sidewall **154**. One or more windows **2028** may be formed in the web **2078**. The window **2028** (for example a relatively large aperture) may be configured to lighten the lower portion **2006**, while maintaining structural integrity of the lower portion **2006**.

[0106] With continued reference to FIGS. **20A** and **20F**, the adjustment apertures **2014b** may be configured to enable the lower portion **2006** to be selectively secured at different angles  $\alpha$  with respect to the carriage **2004**. As shown in FIGS. **20A**, **20B**, **20C**, **20E**, and **20F**, the adjustment apertures **2014b** may be one or more discrete apertures such that the angles  $\alpha$  are predetermined and fixed. The adjustment apertures **2014b** may be arranged in an arcuate or other pattern. In other embodiments, the adjustment aperture **2014b** may be a curved slot that enables continuous adjustment of the angles  $\alpha$ . The lower portion **2006** and the carriage **2004** may be coupled by aligning the pivot aperture **2038a** in the head portion **2032** of the carriage **2004** with the pivot aperture **2038b** of the legs **2076**. A pivot pin **2080** may be received through the pivot aperture

**2038a** and the pivot aperture **2038b**. In some embodiments, a single pivot pin **2080** may extend through the width of the body of the head portion **2032**. In other embodiments, two pivot pins **2080**, one at each end of the head portion, maybe used to couple the respective legs **2076** to the head portion **2032**. The pivot pin **2080** may be a bolts, crew or the like, or may be a pin fixed to or formed with the lower portion **2006**. The adjustment aperture **2014a** may be aligned with one of the adjustment apertures **2014b** and secured with a lock pin **2030**. As with the pivot pin **2080**, the lock pin **2030** may be a single lock pin that extends through substantially a width of the head portion **2032**, or may be two lock pins **2030**, one at each end of the head portion. The adjustment aperture **2014b** may be selected to achieve a desired angle between the carriage **2004** and the lower portion **2006**.

[0107] As shown in FIG. **20B** and FIG. **20G**, the security aperture **2022** may be positioned near a fastener connecting parts of the mounting assembly together in order to obstruct the removal of the fastener, and deter theft. In FIG. **20A**, a security aperture **2022** is adjacent the aperture **2018** in the carriage **2004**, and another security aperture **2022** is positioned on the leg **2076** near adjustment aperture **2014b**. The security apertures **2022** may be configured to receive a locking mechanism **2072** to reduce or prevent tampering with the mounting assembly **2000**. In one example, such as the security aperture **2022** near the adjustment aperture **2014b**, the security aperture may have a central portion **2026** (for example a round central portion **2026**). The security aperture **2022** may have one or more keyways **2024** extending outward from the central portion **2026**. The keyways **2024** may receive a locking mechanism such as a key lock and prevent or limit rotation of the locking mechanism when in the locked position. As shown in FIG. **20G**, with the security aperture **2022** positioned near aperture **2018**, a security cover **2074** may be secured via the locking mechanism **2072** over one or more fasteners such as the fastener **2016** to reduce or prevent tampering with, or theft of, the mounting assembly **2000**. Multiple portions of the mounting assembly **2000** may be secured to one another with the locking mechanism **2072** and security cover **2074**.

[0108] With reference to FIG. **21**, a mounting assembly **2100** is disclosed. The mounting assembly **2100** may be similar to the mounting assembly **2000** in many aspects and may have similar benefits. For example, the mounting assembly **2100** may have a lower portion **2006** and a carriage assembly **2034** as described herein. The carriage assembly **2034** may include a carriage **2004** and a bracket **2102** as described above, except the bracket **2102** may not include a clamp structure for engaging the rim **256** of the sidewall **254**. Instead, the bracket **2102** includes the upper portion **2040** that slidably receives the carriage **2004** as described with respect to FIGS. **20D** above. The adjustable interface between the carriage and the bracket **2102** may be the same or similar as that described with respect to FIGS. **20A-20G**. This example of the bracketed **2102** may be useful with some vehicles, such a light-weight pickup trucks which sometimes do not have a traditional rim **256** the sidewall **254** of the truck bed **252**, and may not be suitable for use with a mounting assembly **2000** that includes a clamp structure. Thus, a mounting assembly **2100** may use one or more fasteners **2104** (e.g., countersink screws, pins, rivets, etc.) received in apertures **2042** formed in the bracket **2102**. The fasteners **2104** may be directly coupled to the vehicle such as by direct threading into the body of the vehicle, such as the rim **256**, or may be coupled via a backing nut, or the like.

[0109] With reference to FIG. **22A**, a mounting assembly **2200** is disclosed. The mounting assembly **2200** may be similar to the mounting assembly **2000** and the mounting assembly **2200** in many aspects and may have similar benefits. For example, the mounting assembly **2200** may have a lower portion **2006** and a carriage assembly **2034** as described herein. The carriage assembly **2034** may include a carriage **2004** and a bracket **2102** as described above, except the bracket **2102** may not include a clamp structure for engaging the rim **256** of the sidewall **254**. The carriage **2004** may interface (for example, slidably via a track **2070**) with a bracket **2202** couplable to the vehicle. The bracket **2202** may have a substantially planar upper portion **2210**. An upper web portion **2204**

may extend from the upper portion **2210** at an angle therefrom (for example approximately 90°) from the upper portion **2040**. A second mid-web portion **2206** may extend back toward the vehicle from the upper web portion **2204**. The second mid-web portion **2206** may be disposed at an angle with respect to the upper web portion **2204**, for example at about 135°. A mounting flange **2208** may extend from the angled portion **2206**, such as in a downward direction. The mounting flange **2208** may be positioned against or adjacent to the sidewall **254** of the truck bed **252** in order to be attached thereto by fasteners. The mounting flange **2208**, in one example, may include one or more apertures **2062** configured to receive fasteners **2064** adapted to secure the mounting flange **2208** to the sidewall **254** of the vehicle. As with the mounting assembly **2100**, the mounting assembly **2200** may not need to be secured to the rim **256** of the sidewall **254**. The mounting assembly **2200** may have the advantage of being compatible with vehicles such as light-weight pickup trucks that do not have a traditional rim on the bed.

[0110] Although the present invention has been described with a certain degree of particularity, it is understood the disclosure has been made by way of example, and changes in detail or structure may be made without departing from the spirit of the invention as defined in the appended claims.

[0111] The description of certain embodiments included herein is merely exemplary in nature and is in no way intended to limit the scope of the disclosure or its applications or uses. In the included detailed description of embodiments of the present systems and methods, reference is made to the accompanying drawings which form a part hereof, and which are shown by way of illustration specific to embodiments in which the described systems and methods may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice presently disclosed systems and methods, and it is to be understood that other embodiments may be utilized, and that structural and logical changes may be made without departing from the spirit and scope of the disclosure. Moreover, for the purpose of clarity, detailed descriptions of certain features will not be discussed when they would be apparent to those with skill in the art so as not to obscure the description of embodiments of the disclosure. The included detailed description is therefore not to be taken in a limiting sense, and the scope of the disclosure is defined only by the appended claims.

[0112] From the foregoing it will be appreciated that, although specific embodiments of the invention have been described herein for purposes of illustration, various modifications may be made without deviating from the spirit and scope of the invention.

[0113] The particulars shown herein are by way of example and for purposes of illustrative discussion of the preferred embodiments of the present invention only and are presented in the cause of providing what is believed to be the most useful and readily understood description of the principles and conceptual aspects of various embodiments of the invention. In this regard, no attempt is made to show structural details of the invention in more detail than is necessary for the fundamental understanding of the invention, the description taken with the drawings and/or examples making apparent to those skilled in the art how the several forms of the invention may be embodied in practice.

[0114] As used herein and unless otherwise indicated, the terms “a” and “an” are taken to mean “one”, “at least one” or “one or more”. Unless otherwise required by context, singular terms used herein shall include pluralities and plural terms shall include the singular.

[0115] Unless the context clearly requires otherwise, throughout the description and the claims, the words ‘comprise’, ‘comprising’, and the like are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to”.

Words using the singular or plural number also include the plural and singular number, respectively. Additionally, the words “herein,” “above,” and “below” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of the application.

[0116] Of course, it is to be appreciated that any one of the examples, embodiments or processes described herein may be combined with one or more other examples, embodiments and/or

processes or be separated and/or performed amongst separate devices or device portions in accordance with the present systems, devices and methods.

[0117] Finally, the above discussion is intended to be merely illustrative of the present system and should not be construed as limiting the appended claims to any particular embodiment or group of embodiments. Thus, while the present system has been described in particular detail with reference to exemplary embodiments, it should also be appreciated that numerous modifications and alternative embodiments may be devised by those having ordinary skill in the art without departing from the broader and intended spirit and scope of the present system as set forth in the claims that follow. Accordingly, the specification and drawings are to be regarded in an illustrative manner and are not intended to limit the scope of the appended claims.

## Claims

1. A mounting assembly for removably attaching components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting assembly comprising: a first bracket configured to be secured to the sidewall and to support a load at an engagement location outside the bed portion and at or below the top rim of the sidewall, the first bracket comprising: a top portion defining at least one elongated track, and a lower portion extending downward from the top portion and defining the engagement location to support the load; and a second bracket configured to directly couple to the sidewall and comprising: an upper portion, and a descending portion configured to couple with a feature of the sidewall; and wherein the top portion of the first bracket is coupled to the upper portion of the second bracket, the first bracket is selectively movable relative to the second bracket along the at least one elongated track.
2. The mounting assembly of claim 1, wherein: the at least one elongated track is defined by: a flange extending along a portion of an edge of the top portion of the first bracket, and a rail extending along a corresponding edge of the upper portion of the second bracket, and the rail and the flange are movably coupled.
3. The mounting assembly of claim 2, wherein: the flange is spaced from a lower face of the top portion and defines a gap, and the rail is received in the gap.
4. The mounting assembly of claim 1, wherein: the top portion defines at least a first aperture, the upper portion defines at least a second aperture, and the first aperture and the second aperture are selectively aligned to receive a fastener to secure the first bracket at a position relative to second bracket along the at least one elongated track.
5. The mounting assembly of claim 1, wherein: the lower portion and the top portion are selectively adjustably and pivotally coupled relative to one another.
6. The mounting assembly of claim 1, wherein the first bracket further comprises: a third portion positioned along a side of the top portion and defining an aperture, wherein the lower portion defines a corresponding aperture, and the lower portion is secured to the top portion by the third portion and by a fastener received in the aperture and the corresponding aperture when aligned.
7. The mounting assembly of claim 6, wherein: the third portion defines a first adjustment aperture, the lower portion defines two or more second adjustment apertures spaced apart, and an angle of the lower portion relative to the top portion is selectively adjusted by inserting a second fastener into the first adjustment aperture and one of the two or more second adjustment apertures.
8. The mounting assembly of claim 6, wherein: the lower portion defines at least one lower flange extending along a portion of one opposing side of the lower portion, and the corresponding aperture is defined in the at least one lower flange.
9. The mounting assembly of claim 1, the second bracket further comprising: a receiving block, wherein the receiving block and the descending portion are selectively coupled by a fastener to define a clamp to releasably couple the mounting assembly to the sidewall.
10. The mounting assembly of claim 1, wherein: the descending portion defines an aperture, and a

fastener extends through the aperture into the sidewall to couple the mounting assembly to the sidewall.

**11.** A mounting assembly for removably attaching components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting assembly comprising: a bracket configured to be secured to the sidewall and to support a load at an engagement location outside the bed portion and at or below the top rim of the sidewall, the bracket comprising: a top portion including at least one adjustment aperture, and a lower portion extending downward from the top portion and defining the engagement location to support the load; and a plate at least partially defining an elongated recess and at least one securement aperture, the plate coupled to the sidewall; wherein the top portion of the bracket is at least partially received within the elongated recess, and selectively securing the bracket relative to the sidewall by aligning the at least one adjustment aperture of the top portion with the at least one securement aperture of the plate and positioning a fastener therethrough.

**12.** The mounting assembly of claim 11, wherein: the elongated recess is defined at least in part by a rail extending along an edge of the plate, the top portion defines a flange along an edge of the top portion, and the flange and rail movably engage to define a track.

**13.** The mounting assembly of claim 11, wherein the lower portion defines a U-shaped cross section defining opposing flanges each coupling with the top portion.

**14.** The mounting assembly of claim 11, wherein the top portion includes a head portion, the head portion is defined at opposing edges of the top portion and extending at least in part away from the top portion, and the opposing flanges each couple with a corresponding side of the head portion.

**15.** The mounting assembly of claim 11, wherein the at least one securement aperture is one of a plurality of securement apertures, and the plurality of securement apertures are spaced apart from a location proximal to the bed portion towards the location outside the bed portion.

**16.** The mounting assembly of claim 11, wherein the lower portion is coupled to the top portion at a first aperture, and selectively connected to the top portion at one of at least two adjustment apertures to selectively position the lower portion at an angle relative to the top portion.

**17.** A mounting assembly for removably attaching components to a vehicle having a bed portion, the bed portion including at least one sidewall defining a top rim, the mounting assembly comprising: a first bracket configured to be coupled to the sidewall and to support a load at an engagement location outside the bed and at or below the top rim of the sidewall, the first bracket comprising: a top portion defining at least a portion of at least one track, and a lower portion extending downward from the top portion and defining the engagement location to support the load; and a securement mechanism configured to directly couple to the sidewall and slidably engage at least a portion of the at least one track, the securement mechanism including a descending portion configured to couple with the sidewall.

**18.** The mounting assembly of claim 17, wherein the portion of the at least one track defined by a slot, at least a portion of the securement mechanism is received in the slot.

**19.** The mounting assembly of claim 17, wherein the slot is defined by a flange extending from one of the opposing edges of the top portion and spaced from a lower face of the top portion, and the securement mechanism defines at least one rail movably received in the slot.

**20.** The mounting assembly of claim 17, wherein the descending portion defines a clamp member to releasably couple the securement mechanism to the sidewall.

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